# Configuring the Elastic Stack for Enhanced Monitoring and Security

## Background and Objective

The Elastic Stack, known for its robustness and flexibility, is widely used in various industries for monitoring, searching, analyzing, and visualizing machine-generated data in real time. By configuring Elasticsearch properly, setting up Packetbeat for network data capture, defining precise security rules, and integrating a Windows agent, students will learn how to create a more secure and insightful monitoring environment.
In this lab, participants will deepen their understanding of the Elastic Stack by configuring advanced settings and deploying additional components. Students will engage in hands-on activities that involve setting up Elasticsearch, Packetbeat, security rules, and a Windows agent, aimed at enhancing network monitoring and security analytics capabilities.

Key Components and Configurations
1. Setup Elastic (Elasticsearch)
Students will configure Elasticsearch to handle larger datasets and more complex queries, adjusting settings that influence performance, such as shard allocations and refresh intervals. This setup is crucial for scaling the deployment to meet the needs of enterprise-level data analysis.
2. Setup Packetbeat
Packetbeat will be configured to capture traffic data from the network, providing insights into the transactions that occur across various protocols. This configuration will help in identifying traffic patterns that may signify security threats or operational issues.
3. Setup Security Rules
Security rules will be defined to automate the detection of anomalies and potential security threats. Students will learn how to write effective rules based on the network data captured by Packetbeat, which can trigger alerts in real-time when potential threats are detected.
4. Setup Windows Agent
A Windows agent will be set up to monitor endpoints effectively. This involves configuring the agent to collect logs, metrics, and security data from a Windows system, which are essential for comprehensive security analysis and incident response.

By the end of this lab, students will have a sophisticated Elastic Stack environment configured for advanced monitoring and security analysis, equipping them with the necessary skills to manage and analyze data in a secure and efficient manner.

# Submission:

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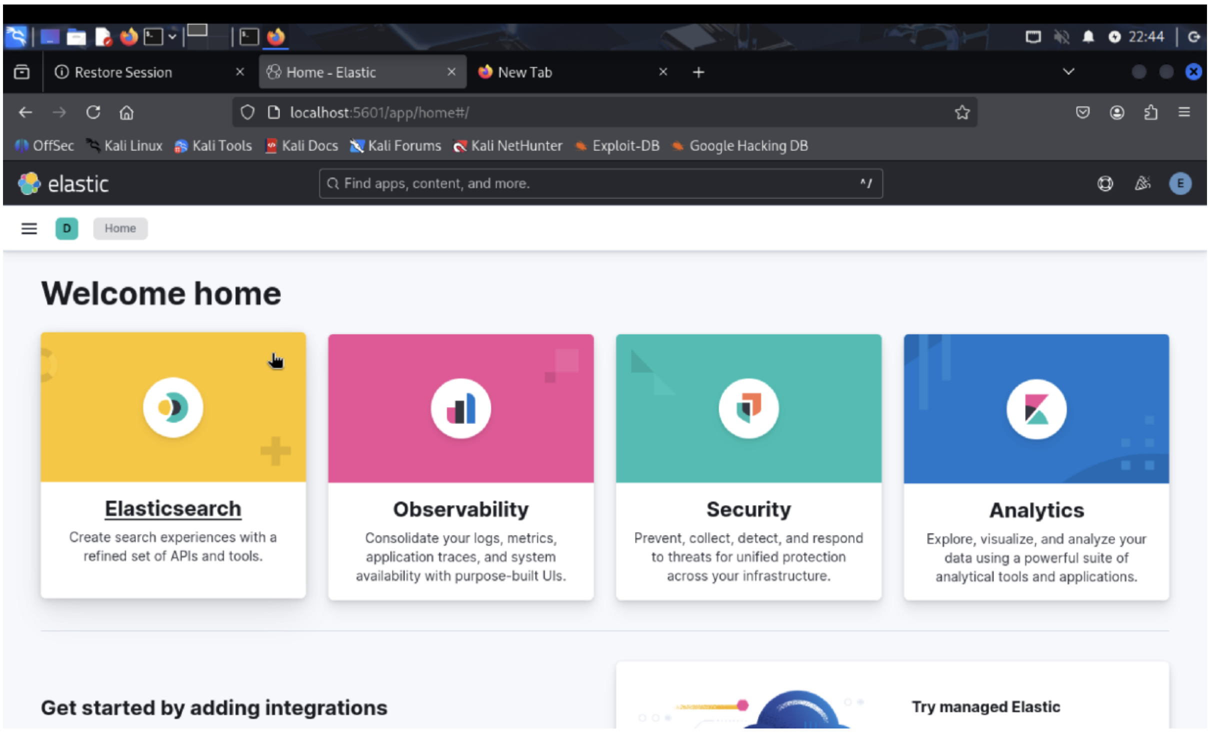
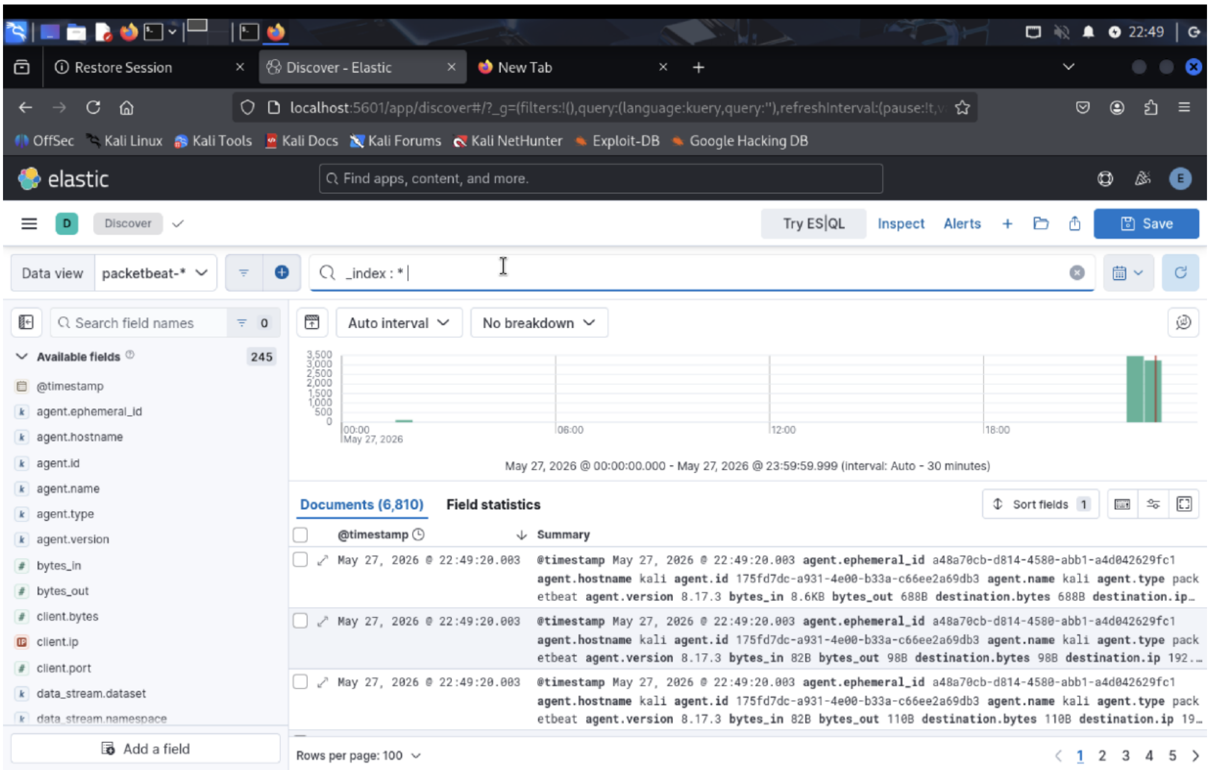
## 1. Packetbeat Dashboard Setup

**1.1** Launch Packetbeat and load default dashboards. Provide a screenshot of the list of dashboards in Kibana.



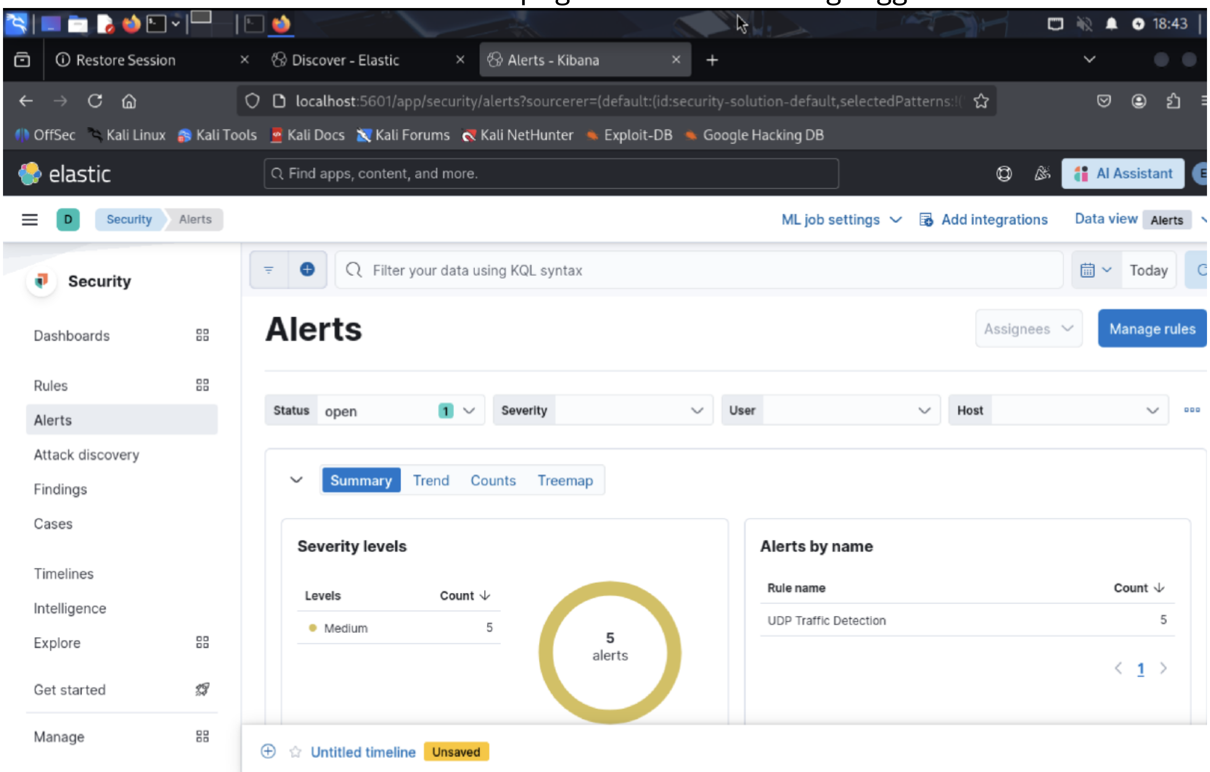

## 2. Discover View – Packetbeat Data

**2.1** Open Kibana Discover tab and select the `packetbeat-*` index pattern. Provide a screenshot showing the latest network logs.

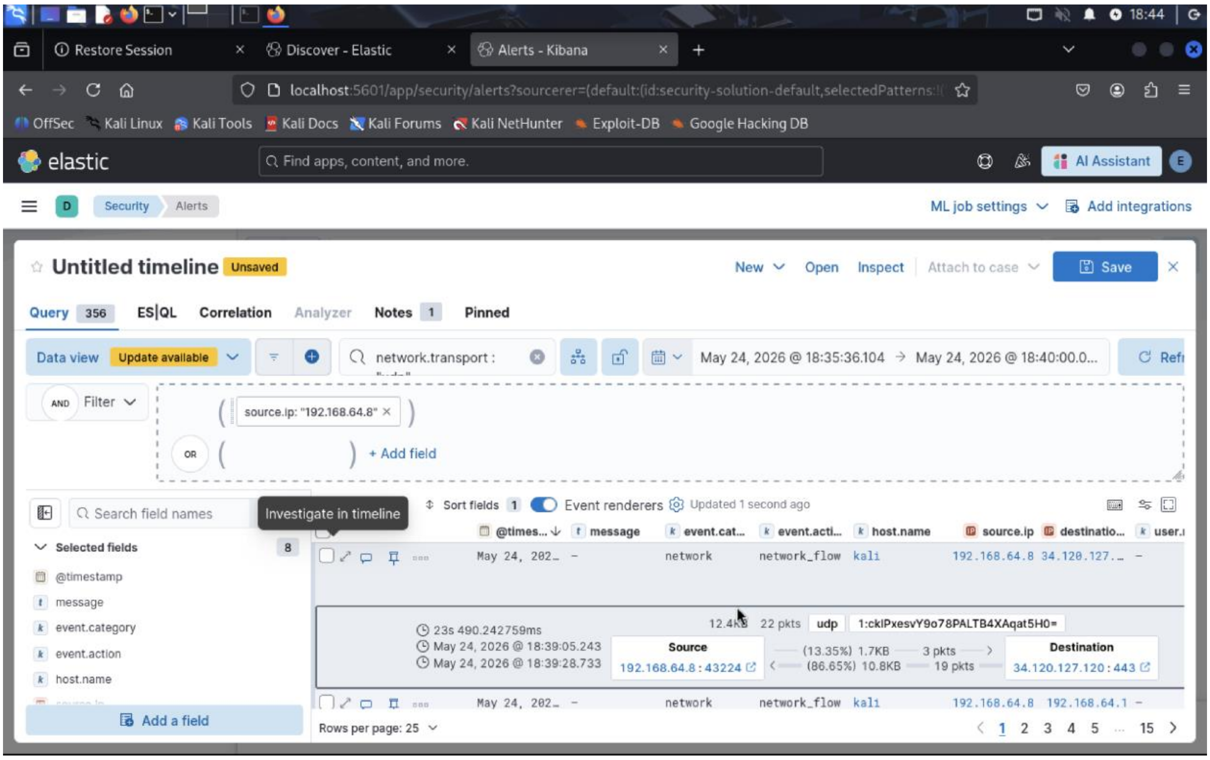



## 3. Security Alerts and Timeline

**3.1** Provide a screenshot of the Alerts page in Kibana showing triggered alerts.




**3.2** Select any alert and capture the event timeline. Provide a screenshot of the alert's timeline view.

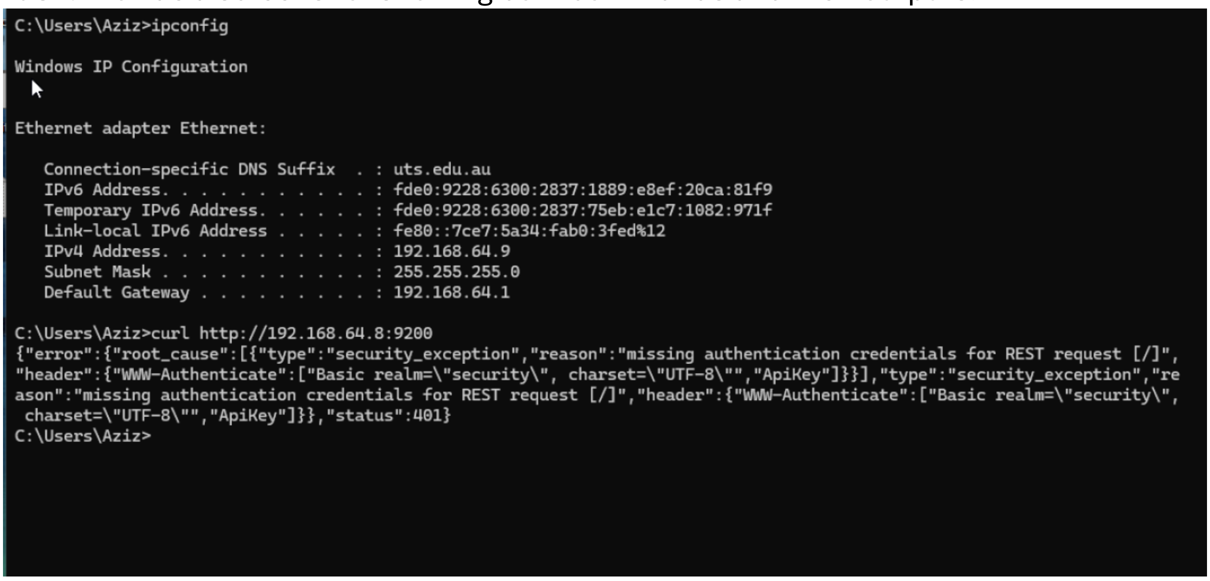



## 4. Windows VM: Elastic Agent Connectivity

**4.1** After setting up the Elastic Agent on Windows VM, open a terminal and run:

```
ipconfig
curl http://[your Elastic host IP address]:9200
```

**Task:** Provide a screenshot showing both commands and their outputs.



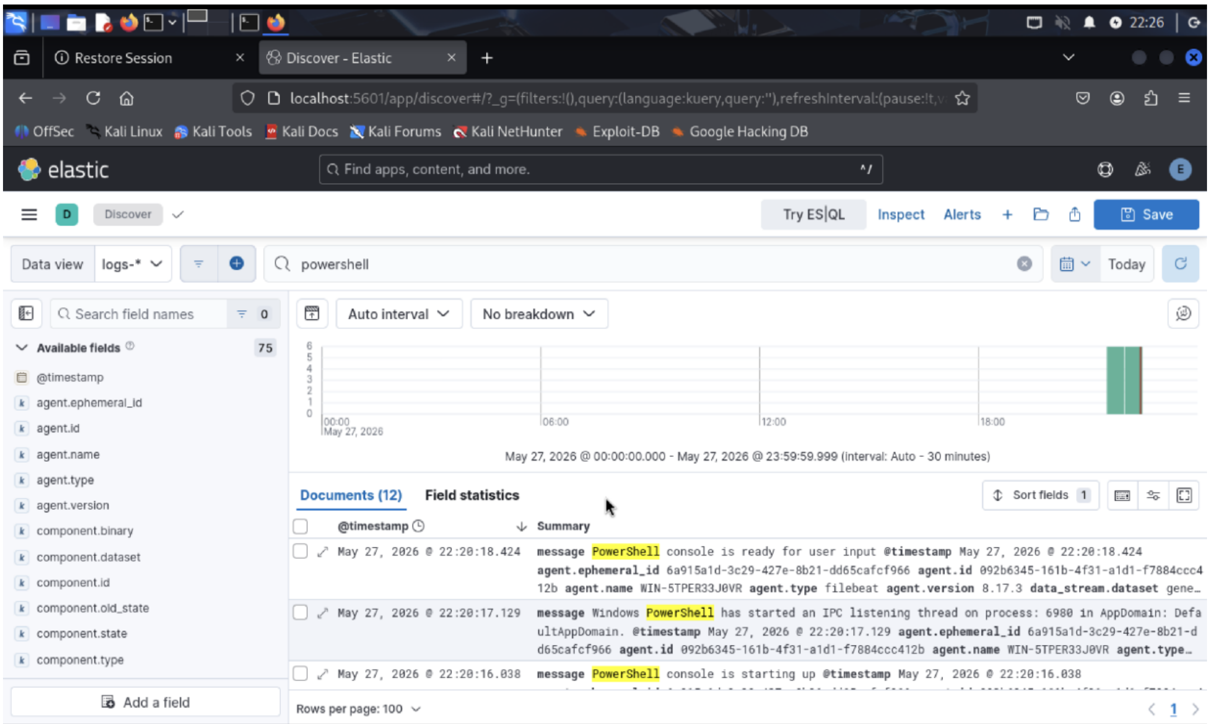
```
C:\Users\Aziz>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : uts.edu.au
   IPv6 Address. . . . . . . . . . . : fde0:9228:6300:2837:1889:e8ef:20ca:81f9
   Temporary IPv6 Address. . . . . . : fde0:9228:6300:2837:75eb:e1c7:1082:971f
   Link-local IPv6 Address . . . . . : fe80::7ce7:5a34:fab0:3fed%12
   IPv4 Address. . . . . . . . . . . : 192.168.64.9
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.64.1

C:\Users\Aziz>curl http://192.168.64.8:9200
{"error":{"root_cause":[{"type":"security_exception","reason":"missing authentication credentials for REST request [/]",
"header":{"WWW-Authenticate":["Basic realm=\"security\", charset=\"UTF-8\"","ApiKey"]}}],"type":"security_exception","re
ason":"missing authentication credentials for REST request [/]","header":{"WWW-Authenticate":["Basic realm=\"security\",
 charset=\"UTF-8\"","ApiKey"]}},"status":401}
C:\Users\Aziz>
```

**4.2** Explain the result of the `curl` command. What does the response indicate about the network connectivity?

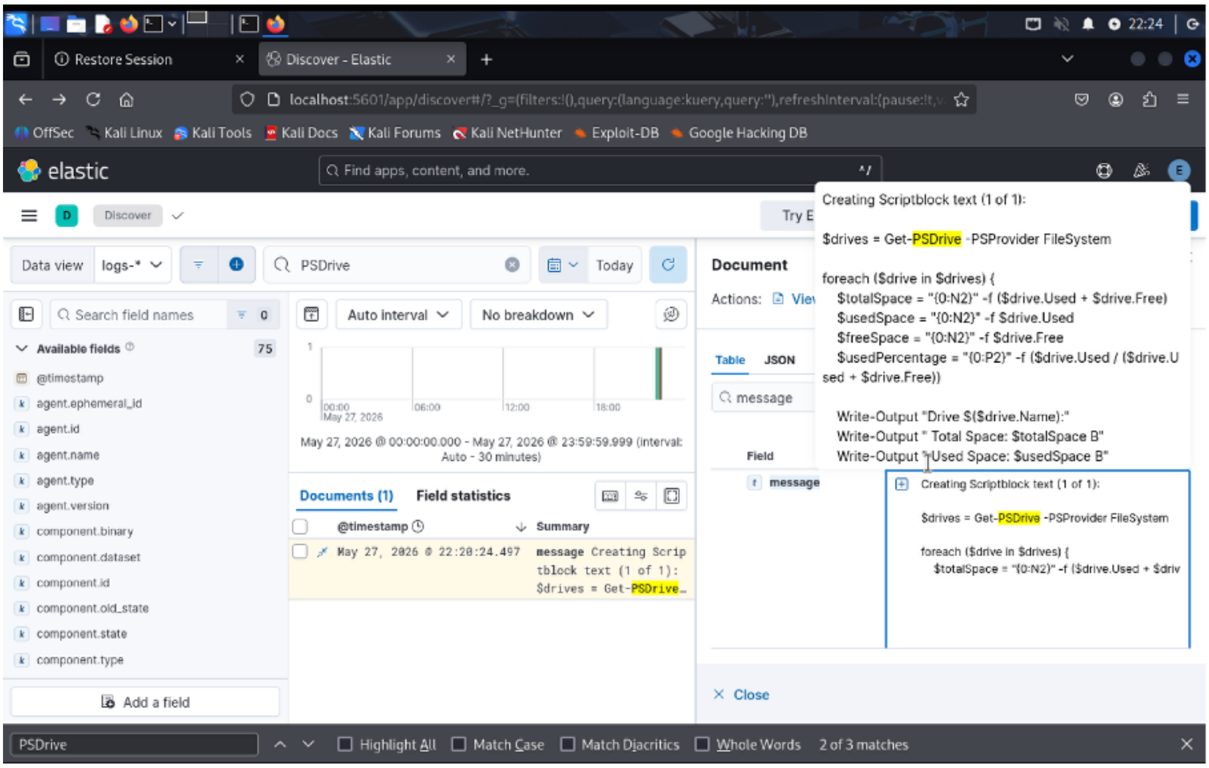
The curl worked and reached the Elasticsearch server, so network connectivity is fine. But it returned a 401 error because no credentials were provided meaning auth is required to actually access it.

## 5. PowerShell Script Execution Logging

**5.1** Execute a PowerShell script (e.g., `getDrive.ps1`) and provide a screenshot of the script execution appearing in Kibana logs.




**5.2** After enabling the log generation in detail, provide detailed logs of script execution with script contents.




## 6. Elastic GenAI for Security

Read the Elastic AI for Security in reading documents or view [https://www.elastic.co/guide/en/security/current/ai-for-security.html](https://www.elastic.co/guide/en/security/current/ai-for-security.html)

Discuss how GenAI technology can benefit threat hunting

GenAI can really boost threat hunting by doing a lot of the heavy lifting analysts normally do manually. Tools like Attack Discovery use LLMs to automatically scan through alerts, find connections between them, and map them to the MITRE ATT&CK framework, so instead of reviewing hundreds of alerts one by one, the AI surfaces what actually matters. The AI Assistant also helps by letting analysts ask questions in plain English, generate queries, and get explanations during investigations. This speeds up the whole process and reduces the time it takes to respond to threats. Basically, GenAI helps threat hunters focus on the real threats instead of getting buried in noise.

## 7. Troubleshooting (If you are facing any)

The main issue I faced was connecting the Windows VM to Elastic (Task 5). The Elastic Agent wasn't communicating with the server, and after spending around 1.5 hours debugging, I found the issue which was the elastic-agent yml file on the Windows machine had incorrect settings. After fixing those, the agent connected successfully and everything worked as expected